

Java Basics – Command Line Arguments, Prime Number and Palindrome

1. Question

Write a Java program to demonstrate Command Line Arguments, check whether a given number is Prime or not, and check whether a given number is Palindrome or not using methods.

2. Steps / Procedure

1. Start the program.
2. Import java.util.Scanner.
3. Create a class named JavaBasics.
4. Create the isPrime() method to check whether a number is prime.
5. Create the isPalindrome() method to check whether a number is a palindrome.
6. In main(), display the command-line arguments.
7. Read a number using Scanner.
8. Call isPrime() and display the result.
9. Read another number.
10. Call isPalindrome() and display the result.
11. Close the Scanner.
12. Stop the program.

3. Algorithm

Prime Number Algorithm

1. Read integer n.
2. If $n \leq 1$, return false.
3. Set $i = 2$.
4. Repeat while $i \leq n/2$.
5. If $n \% i == 0$, return false.
6. Increment i.
7. If no divisor is found, return true.

Palindrome Algorithm

1. Read integer n.
2. Store the original value in original.
3. Set reverse = 0.
4. Extract the last digit using $n \% 10$.
5. Add the digit to reverse.
6. Remove the last digit using $n / 10$.
7. Repeat until $n = 0$.
8. Compare original with reverse.
9. If equal, the number is a palindrome.
10. Otherwise, it is not a palindrome.

4. UML Class Diagram

JavaBasics
+ isPrime(n : int) : boolean + isPalindrome(n : int) : boolean + main(args : String[]) : void

5. Source Code

```
import java.util.Scanner;

public class JavaBasics {

    // Method to check Prime Number
    static boolean isPrime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }

        return true;
    }

    // Method to check Palindrome
    static boolean isPalindrome(int n) {
        int original = n;
        int reverse = 0;

        while (n != 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }

        return original == reverse;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Part 1: Command Line Arguments
        System.out.println("===== COMMAND LINE ARGUMENTS =====");

        if (args.length == 0) {
            System.out.println("No command line arguments provided.");
        } else {
            for (int i = 0; i < args.length; i++) {
                System.out.println("Argument " + (i + 1) + ": " + args[i]);
            }
        }

        // Part 2: Prime Number
        System.out.println("\n===== PRIME NUMBER =====");

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        if (isPrime(number))
            System.out.println(number + " is a Prime Number.");
        else
            System.out.println(number + " is not a Prime Number.");
        // Part 3: Palindrome
    }
}
```

```

        System.out.println("\n===== PALINDROME =====");

        System.out.print("Enter a number: ");
        int palindromeNumber = sc.nextInt();

        if (isPalindrome(palindromeNumber))
            System.out.println(palindromeNumber + " is a Palindrome.");
        else
            System.out.println(palindromeNumber + " is not a Palindrome.");

        sc.close();
    }
}

```

6. Output

Run the program with command-line arguments:

Hello CSM

Then give the following inputs:

17

121

Output Screen

```

===== COMMAND LINE ARGUMENTS =====
Argument 1: Hello
Argument 2: CSM

===== PRIME NUMBER =====
Enter a number: 17
17 is a Prime Number.

===== PALINDROME =====
Enter a number: 121
121 is a Palindrome.

```

7. Result

Thus, the Java program was successfully executed to demonstrate command-line arguments, prime number checking, and palindrome number checking using methods.

8. Program Execution Screenshot

8:52

0.12 KB/s VoLTE 4G+ 84%

StudentUtilityProgram [Java Application] C:\Users\togid\Downloads\eclipse-jee-2025-09-R-win32-x86_64\eclipse\plugins\org

===== STUDENT UTILITY PROGRAM =====

1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade Calculation
5. Day of Week
6. Multiplication Table
7. Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
0. Exit

Enter your choice: 1

Enter Student Name: togi dinkar

Enter Roll Number: 259

Enter Marks in Subject 1: 30

Enter Marks in Subject 2: 50

Enter Marks in Subject 3: 60

|

----- Student Details -----

Student Name : togi dinkar

Roll Number : 259

Total Marks : 140

Percentage : 46.67%

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